AI VIRTUAL MOUSE

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***Abstract*—The AI Virtual Mouse is a ground-breaking touchless alternative to traditional computer mouse, focusing on hygiene and convenience in an increasingly health-conscious environment. While wireless mouse eliminate cable-related issues, they still require physical touch, which can be a vector for germ transmission, particularly in workplaces with shared workstations. This cutting-edge technology uses computer vision and machine learning to eliminate the need for touch by capturing and interpreting hand and finger motions in real time via an external or built-in camera on a computer. The AI Virtual Mouse effectively recognizes hand features via deep neural networks, allowing it to execute fundamental mouse tasks such as clicking, scrolling, and pointer navigation without any Interaction between people. Using OpenCV in Python, this concept not only improves user experience by eliminating the need for physical peripherals, but it also improves hygiene by considerably lowering the danger of contamination associated with high-touch surfaces.**

Keywords—Python, virtual mouse, AI, MediaPipe, OpenCV.

I. INTRODUCTION

Mice have long been a staple PC input device, and as we enter this brave new world of AR and near-universal wireless tech, innovative implementations have sprung up to complement these changes. The proposed system virtually functions like a mouse to accept common mouse input commands (eg: click, scroll and move a cursor) via hand gestures and fingertip detection from fingers using a computer's built-in or external webcam. Using computer vision and machine learning, the groundbreaking system eliminates physical touch in favor of a more time-efficient, sanitary, available and smaller pop-up mouse compared to traditional or Bluetooth mice. Built using Python and OpenCV, MediaPipe, Autopy, PyAutogui additional libraries. The concept here has some resemblance, as this solution tracks and interprets your hand movement to let you control the computer screen without using an external hardware (like USB receiver or a batteries). Not only does this improve HumanComputer Interaction (HCI), but also showcases a robust, high fidelity application with low resource consumption and so can run on standard CPU hardware, allowing for real world functionality without high GPU support. The AI Virtual Mouse system effectively converts hand gestures into key strokes allow intuitive hands-free interaction and serves as an example of how computer vision based virtual devices could change the way we interact with technology.

II. LITERATURE REVIEW

Some researchers have investigated various methods for making virtual mouse control more natural, such as recording finger motions while wearing sleeves or employing color-based gesture detection. These techniques aren't always trustworthy, though. Color-based systems, for example, struggle when the software is unable to reliably discern fingertip colors, and wearing gloves can further impair detection accuracy. Although the use of cameras for hand gesture-based control has been the subject of several research, issues like accuracy and consistency still exist[5], [6].

Lee, Liou, and Hsieh [1] introduced one of the earliest methods using a cursor control system based on hand movement recognition. The issue? Because frames had to be stored before processing, it was more slower and made real-time interaction challenging. Later, Liou, Hsieh, and their colleagues attempted to enhance this with a motion history-based gesture detection system [3], but even that had performance concerns.

Quam [2] adopted a different approach, recognizing gestures using a data glove. Although it was effective, its dependence on intricate hand motions rendered it unsuitable for daily usage. Around the same time, Vinay et al. [11] presented an arm gesture-based cursor control system, which was published in the International Journal of Computers and Applications in 2020. Their system gave different tasks to several bands, but one key drawback was the requirement for separate colors for each function, which isn't always practical in real-world situations. Thakur [4] developed a navigation system that combined finger motions and optics in an effort to make things more seamless. Gopnarayan et al. [5] and Patil et al. [6] advanced the concept by using machine learning to operate a virtual mouse. Despite their potential, these methods still had accuracy issues, particularly when exposed to different lighting conditions.

Fan, Zhang, and Bazarevsky [7] developed the MediaPipe architecture to enhance real-time detection, which greatly increased the effectiveness of finger tracking. Likewise, Kornyakov et al. [8] improved the speed and precision of gesture detection by utilizing OpenCV. In their study of human-computer interactions, Asokkumar et al. [9] emphasized the

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significance of accurate motion tracking for a flawless user experience.

Shibly et al. [10] investigated the potential of abstract hand motions for virtual mouse control and even picture manipulation from an application perspective. Vinay [11] extended this approach with an action-oriented pointer command mechanism. The next phase in human-computer interaction may be brain-computer interfaces, according to Katona [12], who examined neural interaction techniques. Dudhane [13] also donated a hand gesture-based cursor control system, which demonstrates how hands-free computing may progress in the near future.

Despite these advances, gesture-based virtual mouse technology still has a ways to go. Real-time responsiveness, precision, and flexibility in many settings continue to be major obstacles, which makes this a fascinating field for further study and development.

Some Applications are:

This focus is intended to improve the capabilities of various programs inside the Intelligence online cursor system, which may be useful in reducing the area occupied by actual mice-based on an emphasis, as well as in cases when the traditional mouse cannot be used. The technology eliminates the need for devices and improves interfaces between humans and computers.

Principal uses:

- During the COVID-19 pandemic, it is unsafe to utilize equipment by contacting them since this might potentially lead to the virus spreading.
- The technology may be used to control automated systems with no the need for additional equipment.
- AI virtual system able to sketch 2D and 3D images through hand movements.
- VR and AR games can be played via a virtual mouse powered by AI so that there is no need for wireless or cable mouse gadgets.
- It enables people with hand problems to perform mouse actions from the computer.
- HCI based systems can be used to run robots in robotics.
- With this proposed method, virtual prototyping in architecture and design may be employed.

III. DESIGN DETAILS

The system will use the webcam's real-time video input and transform it into array form in order to determine the frame's coordinates. Using these coordinates, the computer can accurately find and identify lifted fingers. The system will carry out certain tasks, such as clicking or moving, by determining which finger is up.

Start: Get things started.

Real-Time: camera footage Acquisition: Record live footage from the computer's camera to keep an eye on hand motions.

Find the Hand Landmarks and Coordinates: Find out the precise locations (coordinates) of particular spots (landmarks) on the hand.

Determine a Finger by Examining the Angle and Fingertip: Analyze the location and angle of each finger to determine which is being utilized.

Perform the Function: This function is to perform the related action (such as mouse click, and scroll) according to the gesture that is recognized.

End: Finish the procedure.

Using hand gestures to operate a computer function without a physical mouse is explained in this flowchart , as seen in the figure 1 below

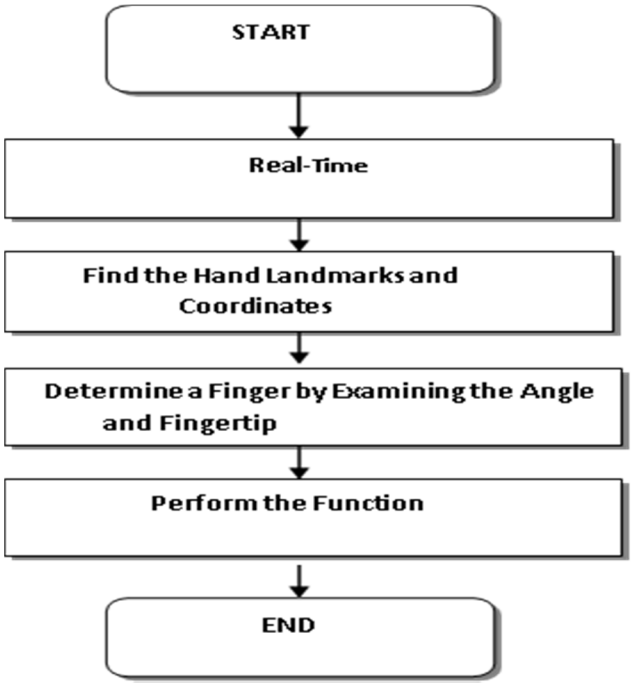


Figure 1 Working of System

IV. METHODOLOGY

Video capture: A camera is used to record live video using the OpenCV module, which then serves as an input for additional processing.

Locate hand landmarks: Using Python libraries like CV2 and MediaPipe we coded the program to locate the hand landmarks. Following hand recognition, it will find 21 spots, as seen in the figure 2.

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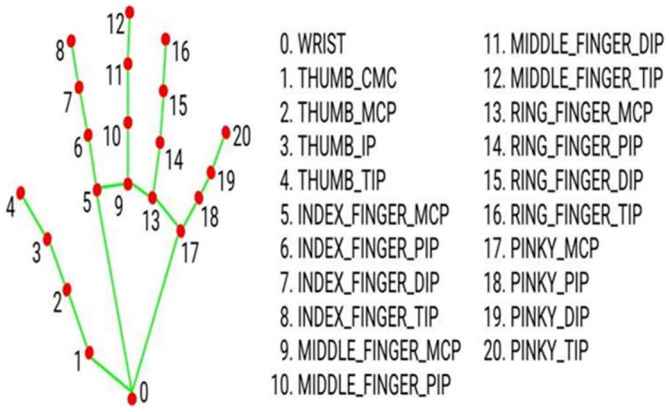


Figure 2 Hand Landmarks

For this one, just have index-and-middle finger at edge tips: To make things simpler, we have incorporated several functions into a module called "HandTrackingModule." This module comprises operations for recognizing hands, identifying fingers, calculating which fingers appear up, and determining finger distances.

Determine which fingers are shown: The software determines which fingers are up.

- **Volume Increase:**If the index, middle, and ring fingers are raised, the system will increase the volume, as seen in the figure 3.

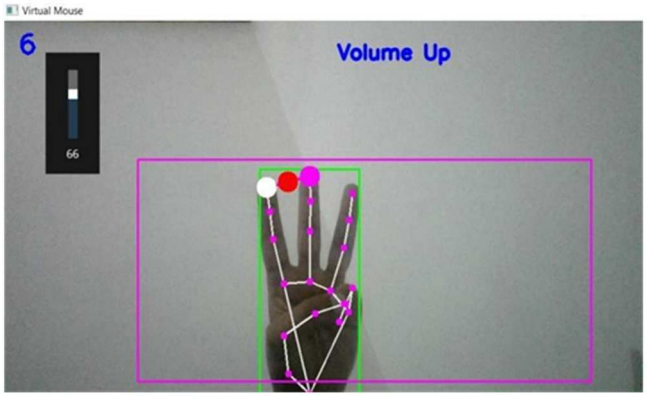


Figure 3 Volume Up

- **Volume Decrease:** If the index, middle, ring, and little fingers are raised, the system will reduce the volume, as seen in the figure 4.

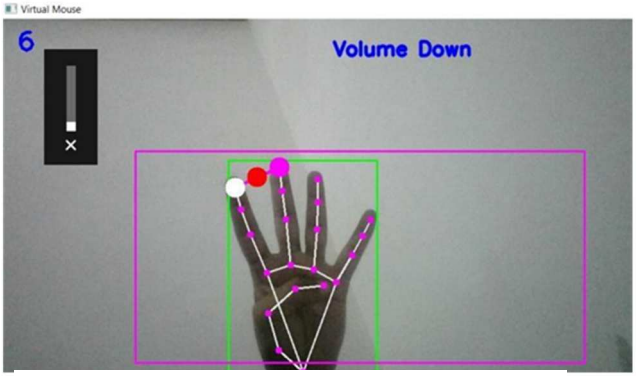


Figure 4 Volume Down

- **Scrolling UP:** If all fingers are raised, the system will scroll upward, as seen in the figure 5.

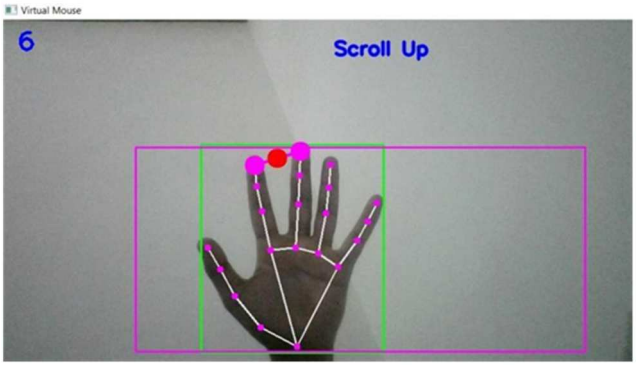


Figure 5 Scroll Up

- **Scrolling Down:** If only the thumb is up, the system will scroll downward, as seen in the figure 6.

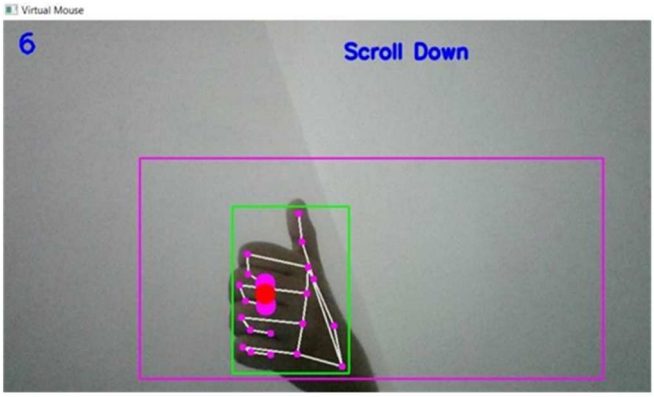


Figure 6 Scroll Down

- **Mouse Movement:** When just the pointer fingertip lifts, the mice is activated will enter moving mode, allowing the user to move the pointer with their finger, as seen in the figure 7.

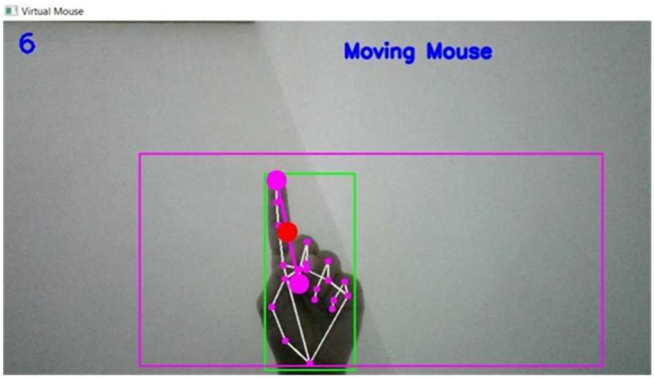


Figure 7 Moving Mouse

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- **Clicking Mode:** The system will go into click mode when the pointer(index) and middle fingertips are lifted simultaneously**,** as seen in the figure 8.

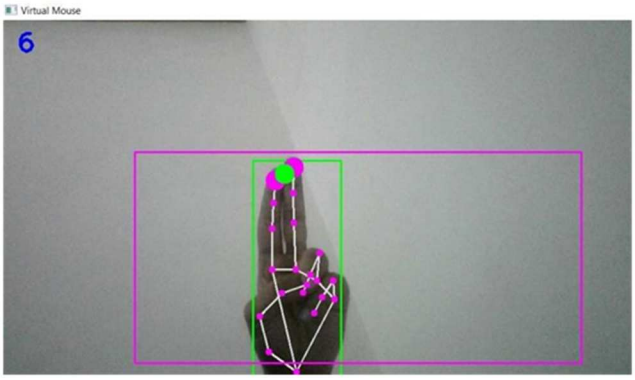


Figure 8 Clicking Mode

Convert the coordinates to obtain the accurate placement: Precise mouse operation depends on the cursor's precise location tracking on the screen.

Put the functions into action: Through the use of coordinates and finger tracking, we are able to digitally operate the mouse without making any physical touch with the device.

Rate of frames: Frame rate enables us to determine whether or not the pointer moves smoothly.

Even out the numbers to prevent jitteriness in the mouse: We included various smoothing techniques to the mouse to make it user-friendly by monitoring changes in frame rate and cursor movement.

Display: We also exhibited camera tracking to demonstrate appropriate implementation, as seen in the figure 9.

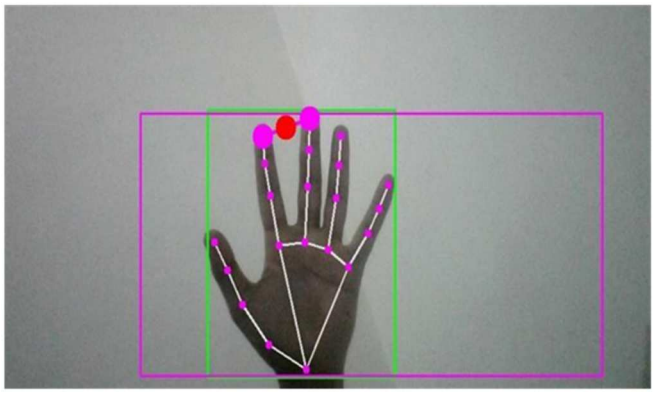


Figure 9 Display

V. RESULTS

Our Python-based Intelligent artificial mice software enables users to manage mice operations via arm gestures, eliminating the requirement for a physical mouse. Using a web cam or integrated camera, our technology tracks hand movements and interprets finger motions as mouse gestures.

The AI Virtual Mouse system, which functions similarly to a genuine computer mice employing machine vision to recognize a hand and fingertip motions for crucial mice operations like clicking, scrolling and pointer navigationg is capable of replacing your traditional Computer mouse. This idea replaces the requirement for a hardware mouse with a built-in or external camera that captures and processes real-time hand gestures to control the computer interface. Constructed using Python and tools such as MediaPipe, PyAutoGUI, AutoPy, and OpenCV, the system precisely recognizes hand landmarks and fingertip motions, allowing for seamless user interaction without requiring hardware contact. This model's high accuracy and real-world applicability make it useful for a variety of situations, including some where a GPU is not required. This touchless interaction method provides a more sanitary and efficient choice, especially in shared or public workplaces, and improves Human-Computer Interaction (HCI) by offering a contactless, gesture-based interface for computer operation.

Details of the user interface: The project will launch a graphical user interface (GUI) window. The primary user interface was this GUI, which includes the Virtual Mouse Exit.
The Figure 10 below displays an active educational message changing while the client chooses decisions.

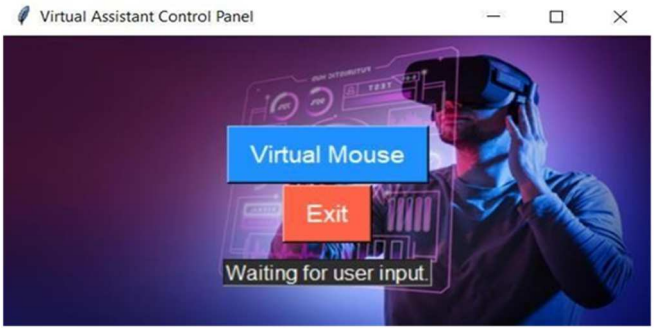


Figure 10 GUI Overview

Client Engagement:

Activating a Digital Controller: When the user hits the "Digital Mice" button, the machine will reply with the following message refers to "Virtual Mouse is operating... This kind of feedback shows that the virtual mouse feature has been effectively enabled.

Exit Option: The Exit button allows clients to stop the program gently, as shown in figure 11 as well.

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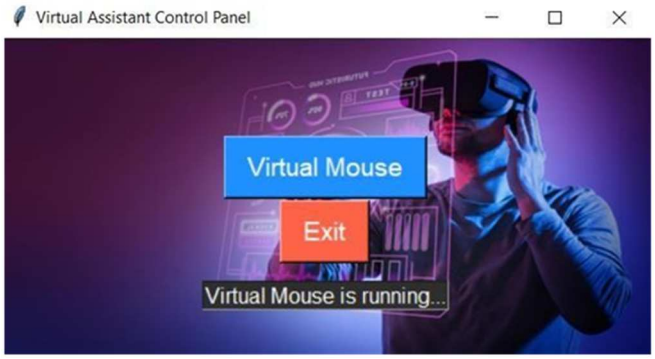


Figure 11 Interaction with Users

Features of an Online Mouse:

Camera inputs: The camera serves as a user input device and is in charge of identifying and following hand motions in order to enable the computer mice feature.

Hand Recognition and Finger Movement: It uses advanced computer vision techniques to monitor hands in real time. For instance, as shown in figure 12, user may map the movement of specific finger locations, such the index finger, to the appropriate mouse function.

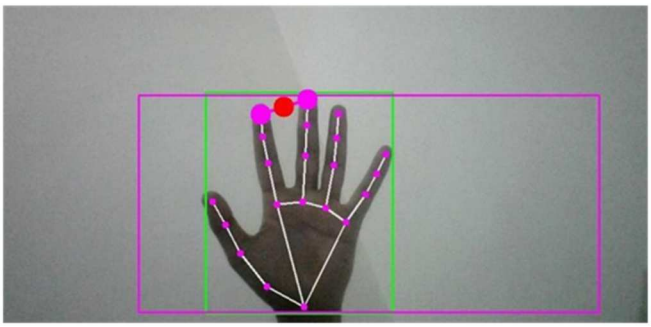


Figure 12 Artificial Cursor Capabilities

For instance, Cursor Movement: The technology uses the pointer finger to move the cursor. Let us show you an example of a mice feature for touch operation replace that can be readily performed utilizing intuitive, fluid, and user-friendly motions, as shown in Figure 13.

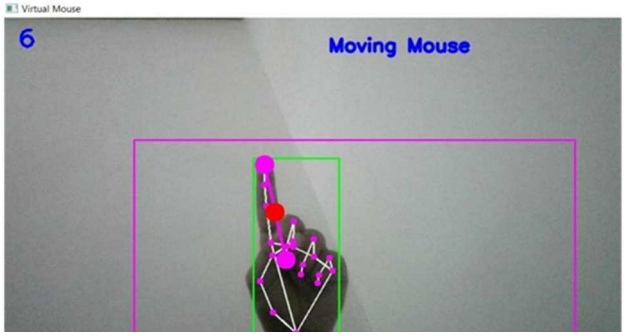


Figure 13 Illustrates the moving click.

VI. Future Scope and Conclusion

Currently, there are some limitations that the AI virtual mouse technology face. It also has a tiny decrease in the accuracy of right- clicking, which may at times affect the precision of particular functions. Moreover, the model struggles with click and drag operations—like text selection—which negatively affects work involving continuous drags. Such problems constitute some of the key restricts connected with Intelligence digital mice, and they will serve as inspiration for future advances in our research. The system can also be further developed in future so as to mimic the keyboard and mouse both virtually. This could create new avenues in Human Computer Interaction(HCI).

It represents the best potential for future development and use. The use of powerful learning techniques will also improve gesture identification, ensuring that accuracy does not suffer in low light or dense backdrops. Adding further platform support, such as macOS, Linux, and mobile compatibility, would increase the system's adaptability and make it available to a larger range of users. Furthermore, the application of such technologies in AR and VR contexts may result in an extra user-friendly and captivating interface, particularly when it comes to gaming or cooperative virtual workplaces. Voice commands and gestures that may be customized for persons with impairments are examples of upcoming accessibility capabilities.

The digital cursor has potential uses in industrial technology and robotics, allowing free of charge and fast operation using gesture- based inputs. A Computer Vision Digital Mice via the Python language and OpenCV is a progress in interaction between humans and computers who avoids contact with input devices and makes certain that there is a means of controlling button motion and go buttons to the application. Using easily accessible gear, such cameras and computer vision, this project shows how simple and user-friendly it can be. The virtual mouse addresses hygiene concerns and can especially helpful in situations where hardware contact must be avoided, such as during a pandemic. Although the system is currently only has basics cursor capabilities, this provides us with a solid foundation upon which to expand in the future.

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